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Cascade recurrent neural network-assisted nonlinear equalization for a 100 Gb/s PAM4 short-reach direct detection system

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We propose a novel, to the best of our knowledge, cascade recurrent neural network (RNN)-based nonlinear equalizer for a pulse amplitude modulation (PAM)4 short-reach direct detection system. A 100 Gb/s PAM4 link is experimentally demonstrated over 15 km standard single-mode fiber (SSMF), using a 16 GHz directly modulated laser (DML) in C-band. The link suffers from strong nonlinear impairments which is mainly induced by the mixture of linear channel effects with square-law detection, the DML frequency chirp, and the device nonlinearity. Experimental results show that the proposed cascade RNN-based equalizer outperforms other feedforward or non-cascade neural network (NN)-based equalizers owing to both its cascade and recurrent structure, showing the great potential to effectively tackle the nonlinear signal distortion. With the aid of a cascade RNN-based equalizer, a bit-error rate (BER) lower than the 7% hard-decision forward error correction (FEC) threshold can be achieved when the receiver power is larger than 5 dBm. Compared with traditional non-cascade NN-based equalizers, the training time could also be reduced by half with the help of the cascade structure. © 2020 Optical Society of America

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Recent years have witnessed the enormous growth of bandwidth-hungry data center interconnects (DCI), which leads to an increasing demand for high capacity short-reach transmission systems. In comparison to coherent detection, which is widely deployed in long-haul fiber transmission, direct detection (DD) is widely employed in short-reach scenarios because of its simple structure and low cost [1]. However, unlike in coherent-detection systems where all the linear optical impairments such as chromatic dispersion (CD) can be compensated for by digital signal processing (DSP) algorithms, the square-law detection in DD systems converts the linear channel effects into nonlinear ones, making it hard to fully compensate for the resulting signal distortion [2,3]. Furthermore, the bandwidth-limited devices, the detrimental frequency chirp of lasers, and the nonlinearity induced by electro-optical

components also contribute to the overall signal distortion, which strongly limits the system achievable capacity [4].

Machine learning (ML) techniques have gained tremendous attraction in optical fiber communication in recent years [5–7]. As a subclass of ML algorithms, a variety of neural networks (NNs) are proposed as the underlying DSP tool to effectively deal with the nonlinear effects in short-reach DD systems, and they outperform the traditional feedforward equalization (FFE)/decision feedback equalization (DFE) or the Volterra series-based equalizer in nonlinear equalization scenarios [8–14]. NN-based equalizers also tend to be more computationally efficient than the commonly used high-order Volterra equalizers. Compared with conventional equalizers, NN-based equalizers can offer better approximation of complex channel effects due to the nonlinear activation functions in each hidden layer, making them suitable to deal with the nonlinear signal distortions in short-reach DD systems. In Refs. [8–10], feedforward NNs (FNNs) of different sizes are employed and proved to be effective nonlinear equalizers for various short-reach transmission schemes. Radial basis function NN (RBF-NN)-based equalizers are proposed in Ref. [11], which could provide more stable results compared with FNN-based ones. Convolutional NN (CNN)-based and recurrent NN (RNN)-based nonlinear equalizers are demonstrated in Refs. [12,13], respectively, both showing good bit-error rate (BER) performance. However, the number of hidden neurons for RBF-NN-based equalizers needs to be sufficiently large to guarantee a relatively good performance [14], and the structures of CNN-based equalizers are much more complicated than those of FNN-based ones. Therefore, RBF-NN and CNN may not be practical considering real-time DSP implementation.

In this Letter, we propose a cascade RNN-based nonlinear equalizer for pulse amplitude modulation (PAM)4 short-reach DD systems which is an extension of our previous work [15]. The system nonlinearity is mainly caused by the mixture of linear impairments (such as CD and bandwidth-limited effects) with square-law detection, the DML frequency chirp, and the device nonlinearity. Enabled by cascade RNN, a 100 Gb/s PAM4 optical link is experimentally demonstrated over 15 km

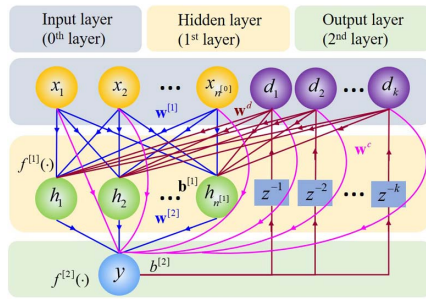


Fig. 1. Schematic of a 2-layer cascade RNN.

standard single-mode fiber (SSMF), using a directly modulated laser (DML) with a 3 dB bandwidth only around 16 GHz. A receiver sensitivity of 5 dBm can be achieved, with 1 dB improvement compared with non-cascade RNN-based equalizers. The computational complexity of cascade RNN is also compared with other NN-based equalizers and it only requires a limited number of additional multiplications on top of the existing FNN structure. What's more, cascade RNN-based equalizers are found to be training-efficient, since the cascade structure helps reduce the epochs for convergence.

The schematic of a 2-layer cascade RNN is shown in Fig. 1, where cascade and recurrent loops are added onto the FNN structure, denoted by the pink and brown arrow, respectively. The cascade networks include connections from the input and every previous layer to the following layers, and there will be only one connection in the 2-layer network, which connects the input layer to the output layer. As illustrated in Fig. 1, the i -th layer contains $n^{[i]}$ neurons, where $i = 0, 1, 2$. To reduce computational complexity, only one output is used ($n^{[2]} = 1$) and the cascade RNN serves as a predictor. For the i -th layer, the weight matrix $\mathbf{w}^{[i]}$ is an $n^{[i]} \times n^{[i-1]}$ matrix containing all the weights connected from the $(i-1)$ -th layer to the i -th layer, the biases are denoted by an $n^{[i]} \times 1$ vector $\mathbf{b}^{[i]}$, and the activation function is denoted by $f^{[i]}(\cdot)$. The recurrent structure is controlled by an $n^{[1]} \times k$ weight matrix $\mathbf{w}^d$ assuming k delays are used in the feedback loop, while the cascade structure is controlled by an $1 \times (n^{[0]} + k)$ weight vector $\mathbf{w}^c$. Taking the received PAM4 symbols as the inputs to the input layer, cascade RNN-based equalization for one symbol recovery refers to one-time forward propagation (FP) of the cascade RNN. Assuming all the weights and biases are obtained through training, the one-time FP step for cascade RNN is given by

$$\mathbf{H}^{[1]} = f^{[1]}([\mathbf{w}^{[1]}, \mathbf{w}^d][\mathbf{X}^T, \mathbf{Y}_d^T]^T + \mathbf{b}^{[1]}), \quad (1)$$

$$y = f^{[2]}([\mathbf{w}^{[2]}, \mathbf{w}^c][\mathbf{H}^{[1]T}, \mathbf{X}^T, \mathbf{Y}_d^T]^T + b^{[2]}), \quad (2)$$

where $\mathbf{H}^{[1]}$ consists of $n^{[1]}$ output values of the hidden layer, $\mathbf{X}$ consists of $n^{[0]}$ inputs, $\mathbf{Y}_d$ contains k delays, and y denotes the output. The feedbacks from the output provide predictions of the previous symbols at low BER, while the cascade loop offers a direct linear mapping between the input and the output; both properties contribute greatly to the overall equalization process and lead to a significant improvement of system BER performance.

The complexity of one-time FP can also be derived from Eqs. (1) and (2), in terms of number of multiplications. Assuming a look-up table is used to deal with the activation function, the number of multiplications needed for the cascade RNN to recover one symbol, denoted N_{mul} , can be expressed as

$$N_{\text{mul}} = (n^{[0]} + k + 1)n^{[1]} + n^{[0]} + k. \quad (3)$$

The experimental setup of a 100 Gb/s PAM4 DML-based short-reach DD optical link with cascade RNN-based equalization is presented in Fig. 2. At the transmitter, the 100 Gb/s random PAM4 symbols are generated by an arbitrary waveform generator (AWG, Keysight M8196A) operating at a sampling rate of 92 GSa/s. A root raised cosine (RRC) filter with roll-off factor of 0.1 is used for Nyquist PAM4 pulse shaping. Then, the signal is amplified by an electronic amplifier with 17 dB gain (SHF 100 BP). The amplified signal is then fed into a DML with an output power of 9.5 dBm and transmitted over a 15 km SSMF. At the receiver side, a variable optical attenuator (VOA) is first used to adjust the received optical power (ROP), and a 43 GHz photodetector (PD, BPDV2150R) is then followed to detect the optical signal. The detected signal is captured by a digital storage oscilloscope (DSO, DSA-X 93304Q) operated at a sampling rate of 80 GSa/s. After resampling, matched RRC filtering and downsampling, the received PAM4 symbols are fed into the cascade RNN-based nonlinear equalizer for symbol decision and BER calculation. The measured power-bias current curve and end-to-end channel response are depicted in the insets of Fig. 2. Shown in inset (a), the output optical power of DML is almost linear to the bias current when the current is larger than 20 mA, and the optimized bias current in our experiment is set as 55 mA. The measured 3 dB bandwidth of the system shown in inset (b) is about 16 GHz, and the measured 10 dB bandwidth is about 20 GHz, which is significantly restricted by the band-limited DML. The limited system bandwidth will bring in significant signal distortion, resulting in poor BER performance.

In the cascade RNN-based DSP module, the current PAM4 symbol, $(n^{[0]} - 1)/2$ past and $(n^{[0]} - 1)/2$ post symbols serve as the cascade RNN inputs, and only one output is the predicted value of the current PAM4 symbol. Compared with the 2-output or 4-output NN structure, only 1 output is employed

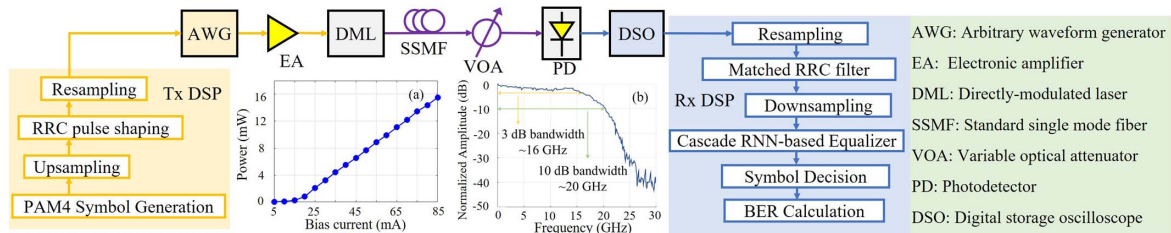


Fig. 2. Experimental setup of a 100 Gb/s PAM4 DML-based short-reach DD system with cascade RNN-based nonlinear equalization. Inset: (a) measured power-bias current curve of the DML, (b) measured end-to-end channel response.

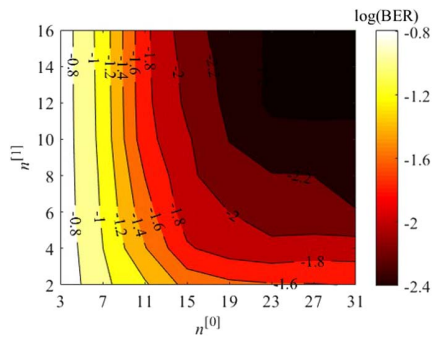


Fig. 3. BER contour map of the 2-layer cascade RNN-based equalizer when the ROP is 5 dBm.

here to reduce computational complexity as much as possible. The number of delays k is optimized and set as 2. For the NN hidden layer, different types of activation functions are tested, and tanh activation function is selected due to its good performance. The pure-linear function is used in the output layer. Twenty thousand random PAM4 symbols are chosen for the training process, 10,000 random PAM4 symbols are used for validation to prevent overfitting, and another 1.2 million random PAM4 symbols are employed for testing, i.e., nonlinear equalization and BER calculation. The symbols in the validation set are different from those in the training set. In the training step, Levenberg-Marquart algorithm (LMA) is selected to carry out efficient update of network parameters, and mean square error (MSE) is adopted as the performance evaluation metric. MSE is commonly used for regression (prediction) and has a strong monotonous relationship with BER. Note that the training process only needs to be performed once for a stationary channel. After the NN is trained, it can be used as a fixed equalizer without retraining. When the channel condition is changed, such as fiber length and bit rate, transfer learning can be applied to enable an efficient transition to new NNs.

Figure 3 shows the BER contour map of the 2-layer cascade RNN-based equalizer with various $n^{[0]}$ and $n^{[1]}$ when the ROP is 5 dBm. As shown in the figure, BER decreases with the increase of $n^{[0]}$ or $n^{[1]}$. However, the BER floor exists for the cascade RNN-based equalizer when $n^{[0]}$ and $n^{[1]}$ are larger than some thresholds. Specifically, no further BER improvement can be observed when $n^{[0]}$ is larger than 23 and $n^{[1]}$ is larger than 12. This is because 23 inputs have already contained enough information of the channel effects and 12 hidden neurons are also sufficient to support nonlinear equalization. When $n^{[0]}$ is beyond 23 and $n^{[1]}$ beyond 12, cascade RNN-based equalizers can achieve BER below the 7% hard-decision forward error correction (FEC) threshold [$\log(\text{BER}) < -2.4$]. Note that compared with some work in Refs. [8–13], a slightly larger network size is needed here since our experiment suffers from more severe impairment due to the high bit rate and band-limited cost-effective DML. As a consequence, $n^{[0]}$ needs to be large to cover sufficient symbols that contribute to inter-symbol interference (ISI), and $n^{[1]}$ should increase correspondingly to guarantee a good performance.

Figure 4 depicts the computational complexity contour map of the 2-layer cascade RNN-based equalizer with various $n^{[0]}$ and $n^{[1]}$, where the selected values of $n^{[0]}$ and $n^{[1]}$ are chosen the same as those in Fig. 3. About 337 multiplications are needed when $n^{[0]}$ equals 23 and $n^{[1]}$ equals 12, which is the minimum

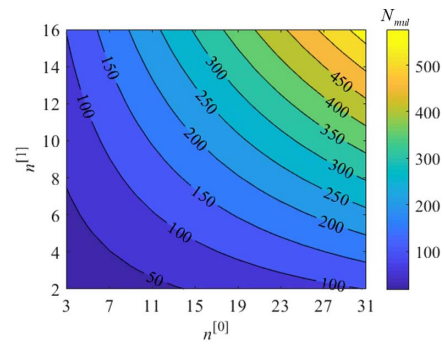


Fig. 4. N_{mul} contour map of the 2-layer cascade RNN-based equalizer.

number of multiplications for a cascade RNN-based equalizer to meet the 7% hard-decision FEC threshold. We also notice that the computational complexity is more sensitive to $n^{[1]}$ compared with $n^{[0]}$, since more weights are connected to the hidden layer. Since a better BER performance normally requires a larger network size, there is always a trade-off between BER and computational complexity, and we can refer to Figs. 3 and 4 to select cascade RNNs that satisfy different requirements.

The relationship between BER and ROP is illustrated in Fig. 5(a), where FNN-, cascade FNN-, RNN-, and cascade RNN-based equalizers are used, all with a size of (23,12). Here we use $(n^{[0]}, n^{[1]})$ to represent the size of 2-layer NNs, and the performance of DD is also shown as a comparison. The cascade RNN-based equalizer outperforms all the other equalizers, where a receiver sensitivity of 5 dBm can be achieved considering 7% hard-decision FEC threshold, showing a 1 dB improvement compared with the second-best non-cascade RNN-based equalizer, and much better than using other NN-based equalizers or DD. We need to mention that when employing 23 inputs and 12 hidden neurons, the 4 types of NN-based equalizers used here all reach their BER floor for a fair comparison. The recurrent structure introduces predictions of past symbols at low BER to deal with the current symbol, while the cascade one provides additional linear paths to directly map the information of inputs to the outputs. Both properties help cascade RNN to achieve the best BER performance.

Figure 5(b) depicts the BER versus ROP with varying receiver bandwidth using cascade RNN-based equalizers with a size of (23,12). The bandwidth is varied by changing the offline digital filter employed after PD. Since the bandwidth of PD is 43 GHz while the 3 dB bandwidth of the signal is only around 16 GHz (limited by DML), we would like to investigate the system performance with lower receiver bandwidths. When the receiver bandwidth is gradually reduced from 24 GHz to 15 GHz, the BER performance drops consequently. The 15 GHz receiver bandwidth shows the worst BER performance, since in this case part of the useful received signal spectrum has been destroyed by the filter. For the 18, 21, and 24 GHz cases, which are larger than the received signal bandwidth, the decrease of BER when shrinking receiver bandwidth indicates that cascade RNN can make use of the signal-to-signal beating term. A receiver sensitivity of 6 dBm can be achieved when the receiver bandwidth is 21 GHz.

In addition to different receiver bandwidths, systems of different bit rates are also demonstrated, and the BER versus bit rate using cascade RNNs of different sizes is illustrated in

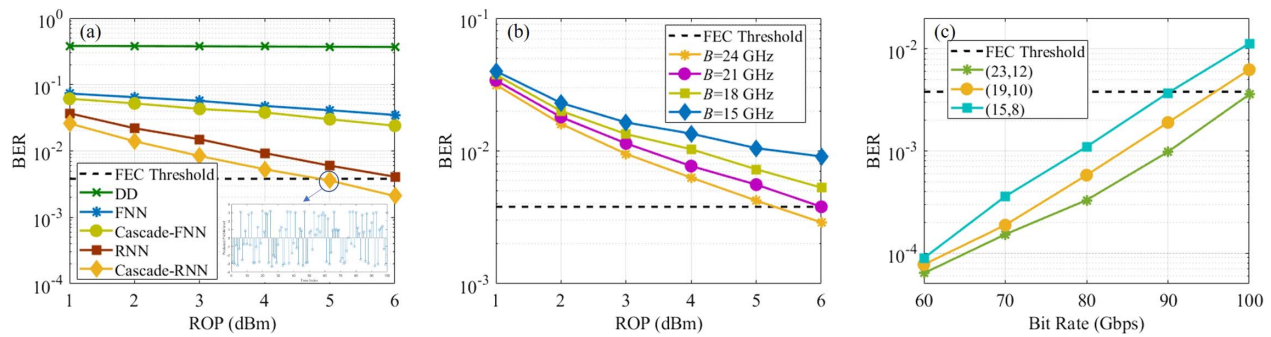


Fig. 5. (a) BER versus ROP using different types of NN-based equalizers with a size of (23,12), inset: predicted PAM4 levels by cascade RNN at a ROP of 5 dBm (100 symbols); (b) BER versus ROP with varying receiver bandwidth using cascade RNN-based equalizers with a size of (23,12); (c) BER versus bit rate using cascade RNNs of different sizes when the ROP is 5 dBm.

Fig. 5(c). The ROP is set as 5 dBm. When the bit rate is 60 Gb/s, the 3 cascade RNNs of different sizes show similar BER performance, since the bit rate is relatively low here, and a cascade RNN with a size of (15,8) could suffice. However, as the bit rate increases, more signal distortions are involved, and therefore, a larger network size is needed to achieve the best BER performance. Nevertheless, the cascade RNN with a size of (15,8) and (19,10) can still support the transmission of a 90 Gb/s and a 95 Gb/s signal, respectively.

Figure 6(a) depicts the number of multiplications N_{mul} of different types of NN-based equalizers with a size of (15,8), (19,10), and (23,12). As shown in the figure, cascade RNNs are the most complicated equalizers because of the cascade and recurrent structure. However, only a limited number of multiplications are needed for cascade RNN compared with FNN of the same size. To be specific, only 25.8%, 20.5%, or 17.0% more of computational complexity are needed for the cascade RNN compared with FNN with a size of (15,8), (19,10), and (23,12), indicating the potential of real-time implementation. Figure 6(b) presents the average training epochs for these NN-based equalizers. The training step terminates until the validation check for each NN fails 3 times. Training steps are performed 20 times, and the average number of epochs are collected to show more reliable results. We can clearly observe that the cascade FNN or cascade RNN reduces the epoch almost by half, compared with the non-cascade counterpart.

In conclusion, we experimentally demonstrated a 100 Gb/s PAM4 short-reach system via a 15 km fiber using a 16 GHz DML in C-band. A cascade RNN-based equalizer is proposed to effectively compensate nonlinear signal distortion, which outperforms non-cascade or non-recurrent NNs. Compared

with FNN with a size of (15,8), (19,10), and (23,12), only 17.0%, 20.5%, or 25.8% additional multiplications are needed for a cascade RNN-based equalizer to recover one symbol. The training epoch can also be reduced by about one-half with the help of the cascade structure.

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Disclosures. The authors declare no conflicts of interest.

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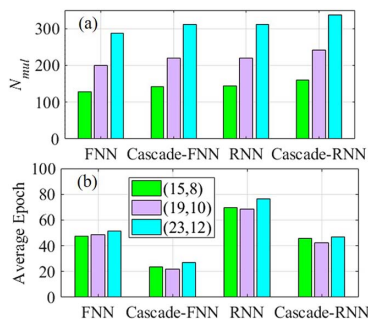


Fig. 6. (a) N_{mul} of different types of NN-based equalizers with different sizes; (b) average training epochs.